PROJECT SYNOPSIS

Title: Automotive Engineering AI Assistant: AUTOSAR HLD Intelligence Platform
Domain: Artificial Intelligence & Machine Learning (AI/ML) / Automotive Systems
Academic Year: 2026

1. ABSTRACT

Modern automotive vehicles incorporate over 100 ECUs governed by AUTOSAR standards. High-Level Design (HLD) documents routinely exceed 100+ pages, detailing complex Software Components (SWCs), communication matrices, and ISO 26262 safety requirements. This project delivers an offline-capable, privacy-preserving AI assistant that ingests, indexes, and queries large automotive HLD documents, guaranteeing 100% citation grounding with exact page numbers.

2. OBJECTIVES

- Automated ingestion of 100+ page automotive HLD PDFs with page-level tracking.
- Grounded RAG Q&A engine delivering sub-second responses with verified page citations.
- Automated reverse-engineering of AUTOSAR SWCs, cycle times, and Port interfaces.
- Automated detection of ISO 26262 ASIL boundary conflicts and timing sampling mismatches.
- Semantic architectural revision comparison between document versions (v1.0 vs v2.0).

3. SYSTEM ARCHITECTURE & MODULES

The system consists of 8 interactive modules: (1) Document Library & Upload, (2) Grounded Q&A Assistant, (3) Technical Search, (4) Architectural Summary, (5) SWC & Port Extraction, (6) Inconsistency & Safety Checks, (7) Revision Comparator, and (8) Analysis Export Center.

4. EVALUATION & RESULTS

- Ingested and benchmarked on an authentic 110-page AUTOSAR Powertrain Master HLD document.
- 20/20 automated Pytest test cases passing (100% pass rate).
- 10/10 RAG evaluation queries achieving 100% retrieval hit rate and 1.4s average latency.

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